Aakash Kulkarni

Software Engineer

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Summary

Software engineer at Cloudvik with 2+ years building backend and cloud systems, focused on GenAI tooling. AWS Certified Solutions Architect — Associate. I like systems that hold up under real users, real failures, and real deadlines.

Skills

Languages	Java, Python, C++, JavaScript, TypeScript, SQL, Bash
Frameworks	Spring Boot, Node.js, FastAPI, React, Next.js, Flutter, Kafka
Cloud & DevOps	AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
Data & AI	PostgreSQL, MySQL, MongoDB, LangChain, RAG, MCP, PyTorch, TensorFlow

Experience

Software Engineer
Sep 2025 – Present

Cloudvik — *Remote*

- Building Spring Boot microservices that orchestrate LangChain extraction pipelines for enterprise document processing.
- Developing REST APIs for a sensor integration platform that ingests real-time telemetry and syncs it with cloud infrastructure.
- Containerizing services with Docker and improving deployment workflows through CI/CD on AWS.
- Leveraging an AI coding harness and MCP tools to accelerate implementation, reviews, and documentation; applying AI governance through scoped tool access, human review gates, and auditable changes.

Graduate Teaching Assistant
Mar 2025 – Jun 2025

Oregon State University — *Corvallis, OR*

- Supported more than 100 students in Parallel Programming, covering multi-threading, synchronization, and concurrent patterns.

Graduate Research Assistant
Apr 2024 – Mar 2025

Oregon State University — *Corvallis, OR*

- Co-developed HeteroBugDetect, a differential fuzzing framework that finds platform-specific bugs in scientific software (LAMMPS).
- Coordinated automated testing workflows on distributed HPC systems, cutting the time to reproduce failures across CPU and GPU builds.

Associate Software Developer
Jun 2021 – Mar 2023

Zemoso Technologies — *Hyderabad, India*

- Built a financial data dashboard for TransUnion with real-time credit trend and risk visualization for enterprise users.
- Optimized database structures for TransUnion's credit record management, improving retrieval speeds for financial modules.
- Implemented OAuth 2.0 authentication workflows to protect sensitive consumer financial data and maintain compliance.

Mobile Application Developer Intern
Jul 2021 – Sep 2021

FindMind Analytics — *Remote*

- Developed ZuPay, a mobile banking app for teenagers with parental controls, in Flutter for iOS and Android.
- Shipped to both app stores, where the app reached 10,000+ active users.

Selected Projects

JAAABS: Job Application Automator 2026

Python, Playwright, LangChain, Next.js, SQLite

- Built a human-in-the-loop studio that discovers companies, scores roles against a binding fact bank, tailors a LaTeX resume per posting, and autofills applications in Chromium.
- Placed human gates at every risky step — login, CAPTCHA, OTP, review, submit — with idempotent run state, an evidence ledger, and a local Next.js dashboard.

The Lazy Radical: Deterministic 4K Motion Pipeline 2026

Python, Playwright, FFmpeg, HTML/CSS

- Replaced timeline editing with a deterministic pipeline: declarative HTML/CSS scenes are frame-stepped by Playwright with a 180° shutter for motion blur, then stitched and mastered by FFmpeg.
- Scenes are versioned and renders are reproducible, so a full 4K video rebuilds from source with one command.

Apparently: Sarcastic Third-Person Biography 2026

SwiftUI, Firebase, Design Systems

- Built an iOS app that turns small logged moments into a monthly chapter written by a sarcastic biographer, backed by a six-theme design system and a fact-bank narrator.
- Structured it as ports and adapters with one composition root and structured logging; the app itself uses no AI at runtime.

AWS Infrastructure Automation Jan 2025

Terraform, AWS, Python, Bash

- Built reusable Terraform modules for VPC, subnets, security groups, EC2, RDS, S3, and Route 53 across dev, staging, and production.
- Automated deployment and monitoring with Python and Bash, and wrote runbooks for scaling and connectivity failures.

Also: **Guitar Hub**, 1st place at OSU Winter Hackathon 2024, and **Soranix**, an Android 16 desktop via proot Debian.

Education

Oregon State University Corvallis, OR

Master of Science in Computer Science • *GPA 3.91/4.0* *Mar 2023 – Jun 2025*

Advisor: Dr. Manish Motwani. Coursework: High Performance Computing, Deep Learning, Software Engineering Methodologies, Parallel Programming.

Jawaharlal Nehru Technological University Hyderabad, India

Bachelor of Technology in Computer Science • *GPA 7.25/10.0* *Jul 2018 – Aug 2022*

Coursework: Data Structures, Algorithms, Operating Systems, DBMS, Networks, Distributed Systems.

Research

LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications — IEEE AIxSE 2025, co-authored with Dr. Manish Motwani. Read the paper.
Combined grammar fuzzing, CPU/GPU differential execution, and LLM triage to surface real defects instead of noise. Found 8 of 20 known bugs plus 2 previously unknown bugs in LAMMPS.

Certifications & Achievements

AWS Certified Solutions Architect — Associate (Jan 2026).
1st Place, OSU Winter Hackathon 2024 — Guitar Hub, an AI-powered guitar tuner and song analysis tool built with OpenAI and Hugging Face.